PRICEOS

AI Revenue Manager

Reservation Agent

Full Build Specification

For Lyzr Studio Development Team

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Launchpad Venture Labs | Dubai, UAE

1. Introduction and Purpose

This document is the authoritative build specification for the PriceOS Reservation Agent. It supersedes the Lyzr Studio Addendum V1.2 and provides complete implementation guidance covering architecture, data models, tool contracts, prompt engineering, voice/call handling, escalation logic, and QA requirements.

The Reservation Agent is a production-grade AI system that handles every job a human reservation agent performs today: inbound guest communication across WhatsApp, email, SMS, and voice calls; proactive pre-arrival and post-departure messaging; operations ticket creation; and revenue-aware upsell at the right moment. It operates inside the PriceOS unified inbox alongside the Revenue (CRO) surface.

> **NOTE** This spec is implementation-ready. Every section maps directly to a Lyzr Studio build task. Developers should read all sections before writing any code.

1.1 Scope of the Reservation Agent

The Reservation Agent covers the full lifecycle of a guest reservation from booking confirmation through post-departure review nudge. It is NOT responsible for dynamic pricing decisions (owned by the Pricing Engine) or operational field work (owned by human ops staff).

In Scope	Out of Scope
All guest-facing communication (pre, during, post stay)	Dynamic pricing decisions
Inbound message triage, classification, and reply drafting	Field operations (cleaning, maintenance work)
Voice call handling (inbound and outbound)	Payment processing or refund execution
Proactive messaging sequences (pre-arrival, mid-stay, post-stay)	Legal representation or dispute resolution
Ops ticket creation with severity and SLA	PMS data entry or reservation amendments
Upsell and early check-in/late checkout offers	New booking creation from scratch
Human escalation routing and approval gates	Owner financial reporting

2. System Architecture

2.1 Agent Position in PriceOS

PriceOS has a Manager-Worker agent architecture. The Reservation Agent is a specialised Worker agent. All agents share a central knowledge base (KB) and communicate via structured event envelopes. The Manager Agent orchestrates task routing and state management.

Layer	Component
Orchestration	Manager Agent (routing, state, escalation decisions)
Revenue	Pricing Engine Worker (CRO surface)
Communication	Reservation Agent Worker (this spec)
Operations	Ops Ticket Worker (receives tickets from Reservation Agent)
Data	Shared KB (listings, reservations, guests, comms, policies)
Channels	WhatsApp, Email, SMS, Voice (Twilio), PMS Webhook

2.2 Comms State Machine

The Reservation Agent reads comms_state at the start of every execution cycle. This is a hard gate: the agent MUST respect this state before any outbound action.

State	Meaning	Agent Behaviour
active	Normal operation	Classify, draft, send, create tickets
paused	Human has taken over thread	Classify, draft only. NEVER send. Escalate if urgent.
syncing	PMS data refreshing	Read-only. Queue actions. Notify Manager Agent.
disabled	Listing deactivated	Do nothing. Log attempted trigger.

> **CRITICAL** MANDATORY: If comms_state != active, the agent must NOT call sendGuestMessage under any circumstances. Violations create guest trust incidents.

2.3 Event Flow

Every agent action produces a typed BaseEnvelope card. No action is performed without a corresponding envelope being written to KB. This enables full auditability and human review.

- Inbound trigger: new message on any channel OR scheduled proactive trigger

- Manager Agent routes trigger to Reservation Agent with context bundle
- Reservation Agent reads comms_state, KB context, and thread history
- Agent classifies intent, sentiment, and confidence
- Agent selects action: draft_reply, escalate, create_ticket, send, close_thread, or upsell
- Action is executed and envelope written to KB
- If human approval required: envelope status = pending_approval
- Human approves or edits via Unified Inbox UI before send

3. Knowledge Base Schema

The KB is the single source of truth for all agent context. Every KB record has an immutable audit trail. Developers must implement all schemas below before building agent tools.

3.1 listing_profile

```
listing_profile:
  listing_id: string (UUID)
  name: string
  address: string
  max_guests: integer
  bedrooms: integer
  bathrooms: integer
  check_in_time: string (HH:MM, 24h)
  check_out_time: string (HH:MM, 24h)
  check_in_method: enum [lockbox, smartlock, key_handover, concierge]
  access_code_location: string (e.g., 'lockbox on left of main door')
  wifi_name: string
  wifi_password: string
  house_rules: string[]
  parking_instructions: string
  amenities: string[]
  emergency_contact: { name, phone }
  owner_id: string
  pm_id: string
```

> **WARNING** Access codes and wifi passwords are read-only for the agent. They may ONLY be sent to verified guests after check-in confirmation, via the approved send_access_details tool. Never interpolate them into free-text replies.

3.2 reservation

```
reservation:
  reservation_id: string (UUID)
  listing_id: string
  guest_id: string
  channel: enum [airbnb, booking_com, direct, vrbo, expedia]
  status: enum [confirmed, checked_in, checked_out, cancelled, no_show]
  arrival_date: date
  arrival_time: string (estimated, HH:MM)
  departure_date: date
  guest_count: integer
  total_amount: decimal
  currency: string
  special_requests: string
  pms_synced_at: datetime
```

3.3 guest_profile

```
guest_profile:
  guest_id: string (UUID)
  name: string
  phone: string
  email: string
  preferred_language: string (ISO 639-1)
  preferred_channel: enum [whatsapp, email, sms, voice]
  vip_flag: boolean
  previous_stays: integer
  notes: string (PM-added, not shared with guest)
```

3.4 guest_thread and guest_message

```
guest_thread:
  thread_id: string (UUID)
  reservation_id: string
  guest_id: string
  channel: enum [whatsapp, email, sms, voice, internal]
  status: enum [open, urgent, pending_approval, closed]
  comms_state: enum [active, paused, syncing, disabled]
  assigned_to: string (agent_id or human_id)
  opened_at: datetime
  last_activity_at: datetime
```

```
guest_message:
  message_id: string (UUID)
  thread_id: string
  direction: enum [inbound, outbound]
  content: string
  handled_by: enum [reservation_agent, human, system]
  intent: string (classified)
  sentiment: enum [positive, neutral, frustrated, angry, distressed]
  confidence: float (0.0-1.0)
  disclose_ai: boolean
  status: enum [draft, pending_approval, sent, failed]
  created_at: datetime
  sent_at: datetime (nullable)
```

3.5 comms_policy

```
comms_policy:
  policy_id: string
  pm_id: string
  tone: enum [formal, friendly, professional]
  languages: string[] (ISO 639-1 codes)
  disclose_ai_default: boolean
  disclose_ai_by_channel: { whatsapp: boolean, email: boolean, sms: boolean, voice:
boolean }
  auto_send_threshold: float (confidence >= this -> auto-send if comms_state=active)
```

```
  escalation_contacts: { name, phone, whatsapp }[]
  upsell_enabled: boolean
  upsell_timing_hours_before_checkin: integer
```

3.6 ops_ticket

```
ops_ticket:
  ticket_id: string (UUID)
  reservation_id: string
  thread_id: string (origin)
  category: enum [maintenance, housekeeping, access, noise, amenity_fault, other]
  description: string
  severity: enum [critical, high, medium, low]
  sla_hours: integer
  status: enum [open, assigned, in_progress, resolved, closed]
  created_by: enum [reservation_agent, human]
  created_at: datetime
  resolved_at: datetime (nullable)
```

4. Tool Contracts

All tools are implemented as Lyzr Studio tool nodes. Every tool must validate inputs, handle errors gracefully, and return a typed response. Developers must not allow agents to call tools with unvalidated parameters.

4.1 Messaging Tools

listThreads

Parameter	Type / Description
status_filter	enum: open | urgent | pending_approval | all
reservation_id	string (optional, filters to single reservation)
limit	integer (default 20, max 100)

Returns: Array of guest_thread records sorted by last_activity_at descending. Includes unread message count per thread.

readThread

Parameter	Type / Description
thread_id	string (required)
include_reservation	boolean (default true, attaches full reservation record)
message_limit	integer (last N messages, default 50)

Returns: Full thread object with messages, reservation details, guest profile, and listing profile.

sendGuestMessage

Parameter	Type / Description
thread_id	string (required)
content	string (required, max 1600 chars for SMS, 4096 for WhatsApp)
disclose_ai	boolean (required, sourced from comms_policy)
approval_required	boolean (set true for any high-risk or low-confidence reply)
channel_override	enum: whatsapp | email | sms (optional, defaults to thread channel)

Side effects: Creates guest_message record (direction=outbound, handled_by=reservation_agent). Updates thread last_activity_at. If approval_required=true, sets message status=pending_approval and notifies human via escalation channel.

> **WARNING** This tool performs a comms_state check internally. If comms_state != active, it rejects the call and returns error code COMMS_STATE_BLOCKED. Always handle this error in the agent flow.

closeThread

Parameter	Type / Description
thread_id	string (required)
reason	string (required, e.g., 'checkout complete', 'issue resolved')
send_farewell	boolean (optional, sends farewell + review nudge if true)

createOpsTicket

Parameter	Type / Description
reservation_id	string (required)
thread_id	string (required, origin thread)
category	enum: maintenance | housekeeping | access | noise | amenity_fault | other
description	string (required, min 20 chars)
severity	enum: critical | high | medium | low
sla_hours	integer (critical=2, high=4, medium=24, low=72)

Returns: ops_ticket record. Automatically notifies ops team via configured channel. Does NOT resolve the guest message thread; agent must also draft a response acknowledging the issue to the guest.

escalateThread

Parameter	Type / Description
thread_id	string (required)
reason	string (required, min 20 chars)
urgency	enum: immediate | high | normal
context_summary	string (required, 3-5 sentence summary for human agent)

draft_reply	string (optional, agent's suggested reply for human to approve or edit)

Side effects: Sets thread status=urgent. Sets comms_state=paused. Sends escalation alert to PM via WhatsApp/Slack with context_summary. If draft_reply provided, creates pending_approval message record.

send_access_details

Parameter	Type / Description
thread_id	string (required)
reservation_id	string (required)
detail_types	enum[]: wifi | access_code | parking | checkin_instructions

Retrieves verified details from listing_profile and formats them into a structured message. Only callable after reservation.status = confirmed and within 24 hours of check-in OR after check-in. Never exposes raw KB fields into free-text generation.

sendUpsellOffer

Parameter	Type / Description
reservation_id	string (required)
offer_type	enum: early_checkin | late_checkout | mid_stay_extension | add_guest
offer_price	decimal (from Pricing Engine)
currency	string
expires_at	datetime

logVoiceCall

Parameter	Type / Description
reservation_id	string (required)
call_sid	string (Twilio call SID)
direction	enum: inbound | outbound
duration_seconds	integer
transcript	string (full call transcript)
summary	string (agent-generated, max 300 chars)

intents_detected	string[] (classified intent list)
outcome	enum: resolved | escalated | ticket_created | no_action
follow_up_required	boolean

5. Voice Call Handling

Voice is a full first-class channel. The Reservation Agent handles inbound and outbound calls using Twilio Voice with real-time transcription. The voice persona must match the comms_policy tone setting and always disclose AI status at the start of the call.

5.1 Inbound Call Flow

1. Twilio receives inbound call and triggers webhook to PriceOS.
2. Manager Agent matches caller phone number to guest_profile.
3. If match found: load reservation, thread, and listing context.
4. If no match: greet as new caller, collect name and reservation reference.
5. Connect to Reservation Agent voice handler with full context bundle.
6. Agent greets guest by name, discloses AI status per comms_policy.
7. Real-time transcription begins. Agent classifies intent every 3-5 sentences.
8. Agent responds using voice-optimised reply templates (short, no bullet points, natural speech patterns).
9. If high-risk intent detected: agent informs guest of transfer, escalates to human.
10. Call ends: logVoiceCall tool writes full record to KB. Thread updated.

5.2 Outbound Call Flow

Outbound calls are triggered by the proactive messaging engine when a guest has not responded to critical pre-arrival messages within the SLA window, or when the PM requests a courtesy call.

11. Proactive engine detects trigger condition (e.g., no response 48h before arrival).
12. Manager Agent evaluates: is outbound call appropriate for this channel preference?
13. If yes: Twilio initiates outbound call with pre-loaded script context.
14. Agent uses scripted opening: purpose of call, AI disclosure, guest name confirmation.
15. If voicemail: agent leaves structured voicemail, logs call, sends follow-up SMS/WhatsApp.
16. If answered: transition to natural conversation using full context bundle.

5.3 Voice Agent Behaviour Rules

- Always disclose AI at the start: 'I'm an AI assistant for [PM Company Name].'
- Never improvise access codes, refund amounts, or policy exceptions verbally.
- If guest asks to speak with a human: acknowledge warmly, initiate transfer within 30 seconds.
- Keep responses under 3 sentences unless guest asks a complex question.
- If connection quality is poor: offer to follow up via WhatsApp/email.
- Log every call regardless of outcome. No silent failures.

5.4 Voice Quality and Latency Requirements

Metric	Target
AI response latency (post-transcription)	< 1.5 seconds
Transcription accuracy target	>= 95% word error rate
Call recording retention	90 days minimum
Escalation transfer time	< 30 seconds from trigger to human connection
Post-call log creation	< 60 seconds after call end

6. Intent Classification System

Every inbound message (text and voice transcript) must be classified before any action is taken. Classification drives routing, escalation decisions, and reply tone. The classification output is stored in the guest_message record.

6.1 Intent Taxonomy

Intent Category	Example Intents	Default Action
Pre-arrival information	check-in time, access code, parking, wifi, directions	Auto-reply if confidence >= 0.85
Arrival logistics	early check-in request, late arrival notification, key collection	Draft reply, may require upsell offer
In-stay support	AC not working, TV issue, missing amenity, noise complaint	Create ops_ticket + draft acknowledgement
Checkout logistics	late checkout request, luggage storage, checkout procedure	Draft reply, may require upsell offer
Payment and billing	invoice request, payment query, pricing question	Draft reply only (no financials disclosed), escalate if dispute
Refund or dispute	refund request, chargeback threat, compensation demand	IMMEDIATE ESCALATION
Safety incident	injury, fire, police involvement, medical emergency	IMMEDIATE ESCALATION + emergency contacts
Legal or media	legal threat, journalist inquiry, social media complaint	IMMEDIATE ESCALATION
Damage claim	broken item, property damage report, deposit dispute	IMMEDIATE ESCALATION
Positive feedback	compliment, thank you, 5-star preview	Auto-reply, flag for review request
Booking inquiry	availability, pricing for dates, new booking request	Draft reply with PMS link or PM contact
General inquiry	local recommendations, restaurant suggestions, transport	Auto-reply using listing KB context

6.2 Confidence Thresholds

Confidence Range	Action
0.90 - 1.00	Auto-send if comms_state = active and intent is not high-risk
0.75 - 0.89	Draft reply, present to PM for quick approval (1-click approve)
0.50 - 0.74	Draft reply with flag, require PM review and possible edit

Below 0.50	Escalate to human. Do not send any draft.

6.3 Sentiment Amplification Rules

Sentiment modulates the confidence threshold. An angry or distressed guest automatically lowers the auto-send threshold by 0.10 and adds a mandatory empathy opener to any draft reply.

Sentiment	Modifier
positive	No change
neutral	No change
frustrated	Lower auto-send threshold by 0.05. Add acknowledgement sentence to reply.
angry	Lower auto-send threshold by 0.10. Require PM review regardless of confidence.
distressed	Lower auto-send threshold by 0.20. Escalate if safety indicators present.

7. Proactive Messaging Sequences

The Reservation Agent executes time-triggered outbound messaging sequences for every reservation. These are the industry-standard sequences every professional PM runs today, delivered automatically and personalised using KB context.

7.1 Booking Confirmation Sequence

Trigger	Message Content
T+0: Booking confirmed	Welcome message. Confirm dates, property name, guest count. Express excitement. Do NOT send access details yet.
T+24h after booking	Introduce the PM company. Share pre-arrival checklist link or FAQ. Invite questions.
T-7 days before arrival (if > 7 days away)	Countdown reminder. Confirm arrival time. Ask if any special requests.

7.2 Pre-Arrival Sequence

Trigger	Message Content
T-48h before check-in	Access details message (triggers send_access_details tool). Wifi, parking, check-in instructions.
T-24h before check-in	Day-before reminder. Confirm estimated arrival time. Share emergency contact.
T-4h before check-in (if upsell_enabled)	Early check-in upsell if unit is available (triggers sendUpsellOffer).

7.3 In-Stay Sequence

Trigger	Message Content
T+4h after check-in	Check-in confirmation message. 'Hope you have settled in well. Anything you need?'
Mid-stay (50% through reservation)	Mid-stay check-in. Local recommendations if applicable. Late checkout upsell if upsell_enabled.

7.4 Checkout and Post-Stay Sequence

Trigger	Message Content

T-24h before checkout	Checkout reminder. Confirm checkout time. Key return instructions.
T-4h before checkout (if upsell_enabled)	Late checkout upsell if PM has availability.
Checkout day T+2h after checkout_time	Farewell message. Confirm successful checkout. Note any items to collect if left behind.
T+24h after checkout	Review request message with direct platform link. Keep short and warm.
T+72h after checkout (if no review)	One gentle follow-up review nudge. Do not send a third reminder.

WARNING All proactive messages must check comms_state before sending. If paused or syncing, queue the message and alert the Manager Agent. Never skip a proactive message silently.

8. Escalation and Human Handover Framework

8.1 High-Risk Intents (Mandatory Immediate Escalation)

The following intents ALWAYS trigger immediate escalation. There is no confidence threshold override. The agent must escalate before drafting any reply to the guest.

High-Risk Intent	Escalation Urgency
Refund request or demand	high
Payment dispute or chargeback threat	immediate
Safety incident (injury, fire, police, medical)	immediate
Legal threat or mention of solicitor / attorney	immediate
Media inquiry or journalist mention	immediate
Property damage claim or dispute	high
Exception to house rules (unless explicitly pre-approved in comms_policy)	high
Guest mentions domestic violence or personal safety risk	immediate
Suspected illegal activity on premises	immediate

8.2 Escalation Execution Steps

17. Detect high-risk intent during classification.
18. Do NOT draft or send any reply to the guest yet.
19. Call escalateThread with urgency level, reason, and context_summary.
20. Send guest an acknowledgement only: 'We have received your message and a member of our team will be in touch shortly.'
21. Set comms_state = paused on the thread.
22. Alert PM via WhatsApp with context_summary and thread link.
23. Optionally include agent draft_reply for PM to approve or rewrite.
24. Log escalation as guest_message (handled_by=reservation_agent, status=sent).

8.3 Human Takeover Protocol

When a human takes over a thread, the following rules apply:

- comms_state is set to paused by the human via the Unified Inbox UI.
- Agent continues to monitor thread and classify inbound messages silently.
- Agent does NOT send any messages while paused.
- Agent may create draft replies visible to the human in the draft card.
- Agent creates ops_tickets for any operational issues raised during human-managed period.
- Human must explicitly resume by setting comms_state = active via UI.
- On resume, agent sends an acknowledgement if the thread was silent for > 2 hours.

9. Reply Generation Guidelines

9.1 Tone and Voice

Replies must match the comms_policy.tone setting. All tone variants share common principles: be specific (use guest name, property name, actual times), be brief (most replies should be 2-4 sentences), and be action-oriented (always tell the guest what happens next).

Tone Setting	Character	Example Opening
formal	Professional, courteous, structured. Full sentences.	Dear [Name], thank you for reaching out regarding your reservation.
friendly	Warm, human, slightly casual. Contractions allowed.	Hi [Name]! Thanks for your message.
professional	Balanced. Warm but efficient. Default for Dubai market.	Hi [Name], thanks for getting in touch.

9.2 Reply Construction Rules

- ALWAYS use the guest's first name in the opening.
- ALWAYS include a specific next step or action at the end.
- NEVER include placeholder text or template variables in a sent message.
- NEVER disclose that the reply was AI-generated unless disclose_ai = true.
- NEVER include pricing, refund amounts, or financial figures.
- NEVER include access codes or wifi passwords in free-text replies. Use send_access_details tool.
- Keep replies under 150 words for WhatsApp and SMS.
- Email replies may be longer but must still have a clear subject line.
- For voice: construct reply as natural spoken sentences. No bullet points. No markdown.

9.3 Language Handling

The agent must reply in the guest's language if it is included in comms_policy.languages. If the detected language is not in the policy list, reply in the primary language (first in the list) and note the limitation to the PM in the escalation notes.

9.4 AI Disclosure

AI disclosure must appear naturally at the end of the first message in a thread, not as a legal disclaimer header. Example: 'This message was sent by an AI assistant on behalf of [PM Company].' Voice calls must disclose at the start of the call.

10. Output Envelopes (BaseEnvelope Cards)

The agent must return structured envelope cards for every action. No unstructured text output is permitted in production. All envelopes are stored in KB and surfaced in the Unified Inbox UI.

10.1 draft_reply

```
{
  "envelope_type": "draft_reply",
  "thread_id": "string",
  "message_id": "string",
  "content": "string",
  "intent": "string",
  "sentiment": "string",
  "confidence": 0.0,
  "requires_approval": true,
  "channel": "string",
  "disclose_ai": true,
  "created_at": "ISO8601"
}
```

10.2 reply_sent

```
{
  "envelope_type": "reply_sent",
  "thread_id": "string",
  "message_id": "string",
  "content": "string",
  "channel": "string",
  "sent_at": "ISO8601"
}
```

10.3 escalation

```
{
  "envelope_type": "escalation",
  "thread_id": "string",
  "intent": "string",
  "urgency": "immediate | high | normal",
  "reason": "string",
  "context_summary": "string",
  "draft_reply": "string (nullable)",
  "alerted_contacts": ["string"],
  "created_at": "ISO8601"
}
```

10.4 ticket_created

```json
{
  "envelope_type": "ticket_created",
  "ticket_id": "string",
  "reservation_id": "string",
  "category": "string",
  "severity": "string",
  "sla_hours": 0,
  "description": "string",
  "created_at": "ISO8601"
}
```

10.5 thread_closed

```json
{
  "envelope_type": "thread_closed",
  "thread_id": "string",
  "reason": "string",
  "farewell_sent": true,
  "closed_at": "ISO8601"
}
```

10.6 upsell_sent

```json
{
  "envelope_type": "upsell_sent",
  "reservation_id": "string",
  "offer_type": "string",
  "offer_price": 0.0,
  "currency": "string",
  "expires_at": "ISO8601",
  "sent_at": "ISO8601"
}
```

11. Lyzr Studio System Prompt

This is the drop-in system prompt for the Reservation Agent in Lyzr Studio. It must be loaded as the agent's base system prompt. Tool definitions, KB connections, and channel integrations are configured separately in the Studio UI.

> **CRITICAL** Do NOT modify this prompt without versioning the change and testing against all QA scenarios in Section 13. Prompt changes have immediate production impact.

```
You are the PriceOS Reservation Agent for a Dubai holiday-home property management
company. You handle all guest communication inside the unified inbox across WhatsApp,
email, SMS, and voice. You are the primary contact for guests from booking
confirmation
through post-departure review.
```

```
IDENTITY AND DISCLOSURE
- You are an AI assistant operated by the PM company.
- When disclose_ai = true: append 'This message was sent by an AI assistant on behalf
  of [PM Company Name].' at the end of the first message in a new thread.
- On voice calls: always open with 'Hi [Name], I am an AI assistant for [PM Company].
  How can I help you today?'
```

```
MANDATORY GATE: COMMS STATE
- Read comms_state at the start of every execution.
- If comms_state != active: DO NOT call sendGuestMessage. Draft only. Escalate if
urgent.
- If comms_state = disabled: do nothing. Log attempted trigger. Exit.
```

```
CLASSIFICATION
- Classify every inbound message: intent + sentiment + confidence (0.0-1.0).
- If confidence < 0.50: escalate immediately. Do not draft a reply.
- Adjust confidence threshold downward by 0.05-0.20 for frustrated/angry/distressed
  sentiment (see sentiment amplification rules).
```

```
REPLY RULES
- Use guest's first name in every opening.
- Reply in guest's detected language if it is in comms_policy.languages.
- Keep replies under 150 words for WhatsApp and SMS.
- End every reply with a specific next step.
- Never include placeholders, template variables, or unresolved tokens.
- Never include access codes, wifi passwords, or financial figures in free-text.
  Use send_access_details tool for credentials.
- Never guess: if unsure about any factual detail, escalate.
```

```
HIGH-RISK INTENTS (ALWAYS ESCALATE BEFORE REPLYING)
- Refund request, payment dispute, chargeback threat
- Safety incident (injury, fire, police, medical emergency)
- Legal threat, solicitor/attorney mention
- Media or journalist inquiry
- Property damage claim or dispute
```

```
- House rules exception request (unless pre-approved in comms_policy)
- Domestic violence or personal safety mention
- Suspected illegal activity
```

```
OPS TICKETS
- Create ops_ticket for every maintenance, housekeeping, or amenity issue.
- Set severity: critical (safety/no access), high (major amenity failure),
  medium (minor inconvenience), low (cosmetic).
- Set sla_hours: critical=2, high=4, medium=24, low=72.
- Always send a guest acknowledgement after creating a ticket.
```

```
PROACTIVE SEQUENCES
- Execute all proactive triggers on schedule (see Section 7).
- If upsell_enabled: send upsell offers at configured timing windows.
- Always check comms_state before any proactive send.
```

```
OUTPUT
- Return BaseEnvelope cards only: draft_reply, reply_sent, escalation,
  ticket_created, thread_closed, upsell_sent.
- No unstructured prose output in production mode.
- Every action must produce an envelope written to KB.
```

12. Unified Inbox UI Requirements

The Unified Inbox is a two-surface app shell: Revenue (CRO) and Guest Inbox (Reservation Agent). This section defines the Reservation Agent surface requirements only.

12.1 Thread List View

- Display open and urgent threads in two tabs.
- Each thread row: guest name, listing name, last message preview (40 chars), time since last activity, unread count badge.
- Urgent threads pinned to top with red indicator.
- Filter by: listing, channel, comms_state, date range.
- Search by guest name, reservation ID, or message content.

12.2 Thread Detail View

- Full message history in chronological order.
- Each message: sender, timestamp, channel icon, AI/human handled badge.
- Inbound messages show: intent tag, sentiment badge, confidence score.
- Pending approval messages show: approve button, edit button, reject button.
- Active ops tickets linked inline with status indicator.

12.3 Draft Reply Card

- Agent draft displayed below thread, clearly labelled 'AI Draft'.
- Editable text field. PM can edit before approving.
- Send button (sends immediately if comms_state = active).
- Escalate button (triggers escalateThread, sets comms_state = paused).
- Discard button (discards draft, does not affect thread state).

12.4 Comms State Control

- Visible toggle in thread header: Active / Paused.
- Changing state requires confirmation modal: 'Pause AI replies for this thread?'
- State change is logged as system guest_message with handled_by=system.

13. QA Test Scenarios

All scenarios below must pass before production deployment. Each scenario maps to a specific agent behaviour requirement. QA engineer must verify both the agent action AND the KB record created.

13.1 Core Flow Tests

Scenario	Expected Behaviour
Guest asks for wifi password via WhatsApp	Agent uses send_access_details tool. Does NOT include password in free-text reply.
Guest requests early check-in, upsell_enabled = true	Agent drafts upsell offer at configured price from Pricing Engine.
comms_state = paused, guest sends message	Agent classifies, drafts reply, does NOT send. Creates escalation if urgent.
Guest reports broken AC mid-stay	Agent creates ops_ticket (category=maintenance, severity=high, sla=4h). Drafts guest acknowledgement.
Inbound message in Arabic, Arabic in comms_policy.languages	Agent replies in Arabic matching the tone setting.
Guest sends late checkout request	Agent checks upsell_enabled. If true: sends upsell offer. If false: checks house rules and drafts polite response.
Guest explicitly asks if speaking to AI	If disclose_ai = true: agent confirms AI status clearly and warmly. If false: agent follows PM disclosure policy.

13.2 Escalation Tests

Scenario	Expected Behaviour
Guest demands refund	IMMEDIATE ESCALATION. Agent sends acknowledgement only. Does not discuss refund amount or policy.
Guest mentions injury on property	IMMEDIATE ESCALATION with urgency=immediate. Emergency contact notified. comms_state = paused.
Guest sends legal threat	IMMEDIATE ESCALATION. No draft reply. PM alerted via WhatsApp within 60 seconds.
Confidence = 0.45 on ambiguous message	Agent escalates. Does not draft reply. Logs classification attempt.
Guest is classified as angry sentiment	Auto-send threshold lowered by 0.10. If resulting threshold not met: draft requires PM approval.

13.3 Voice Call Tests

Scenario	Expected Behaviour
Inbound call, guest matched by phone	Agent greets by name. Discloses AI. Loads full reservation context.
Guest asks for access code on call	Agent verbally confirms access code method but does NOT read code aloud. Says: 'I have sent the full access details to your WhatsApp.'
Guest demands refund on call	Agent acknowledges, informs guest a team member will follow up, transfers call. logVoiceCall writes full transcript.
Outbound call goes to voicemail	Agent leaves structured voicemail. Sends follow-up WhatsApp. logVoiceCall with outcome=no_action.

13.4 Proactive Sequence Tests

Scenario	Expected Behaviour
Booking confirmed T+0	Welcome message sent within 5 minutes of reservation.status = confirmed.
T-48h pre-arrival trigger fires	send_access_details tool called. Access details sent as structured message, not free-text.
comms_state = syncing when proactive trigger fires	Message queued. Not sent. Manager Agent alerted. Sent when comms_state returns to active.
T+24h post-checkout, no review submitted	Review request sent once. No further nudges unless T+72h nudge window reached.

14. Developer Integration Checklist

Complete all items in order. Do not proceed to the next phase until all items in the current phase are verified.

Phase 1: Data Foundation

- KB schemas implemented for all 6 entities (Section 3)
- All schemas have immutable audit trail (created_at, updated_at, updated_by)
- PMS webhook integration live and writing to reservation KB
- guest_profile phone number normalisation to E.164 format
- comms_state machine implemented and enforced at KB level

Phase 2: Tool Implementation

- All 8 tools implemented with input validation (Section 4)
- sendGuestMessage internal comms_state check implemented
- send_access_details tool: verified guest check before releasing credentials
- createOpsTicket: ops team notification configured
- escalateThread: WhatsApp/Slack alert with context_summary verified
- logVoiceCall: connected to Twilio call_sid lookup

Phase 3: Channel Integrations

- WhatsApp Business API: inbound and outbound verified
- Email: SMTP outbound + inbound parsing via webhook
- SMS: Twilio SMS inbound and outbound
- Voice: Twilio Voice inbound webhook, real-time transcription, outbound dialler
- Channel routing: all inbound messages correctly routed to guest_thread by channel

Phase 4: Agent Configuration

- System prompt loaded in Lyzr Studio (Section 11)
- All 8 tool nodes attached to agent
- KB read connections: listing_profile, reservation, guest_profile, guest_thread, comms_policy
- KB write connections: guest_message, ops_ticket
- Classification model tested on 50 sample messages across all intent categories

Phase 5: QA

- All 21 QA scenarios passed (Section 13)

- comms_state gate tested for all 4 states
- High-risk intent escalation tested for all 10 scenarios
- Voice call end-to-end tested (inbound and outbound)
- Proactive sequence timing verified against timezone (Dubai = UTC+4)
- AI disclosure tested for both disclose_ai = true and false

Phase 6: Production Readiness

- Error handling: all tool failures produce meaningful log entries
- Rate limiting: WhatsApp 1000 messages/day Business API limit monitored
- Call recording retention policy configured (90 days)
- PM onboarding: comms_policy configured per property
- Escalation contacts configured per property
- Staging environment smoke test with live Twilio and WhatsApp credentials

15. Revision History

Version	Changes
V1.0	Initial Lyzr Studio Addendum. Unified Inbox UX, basic tool contracts, escalation rules.
V1.2	Added comms_state machine, KB input spec, proactive sequence outline, Studio prompt.
V2.0 (this document)	Full production spec. Added voice/call handling, complete intent taxonomy, sentiment amplification, proactive sequence detail, QA scenarios, developer integration checklist, complete KB schemas, all 8 tool contracts with parameters.

Document prepared by: Launchpad Venture Labs

For Lyzr Studio development team use only.